University of Technology, Jamaica

Faculty of Engineering and Computing - School of Computing and Information Technology

Web Programming

CIT2011

Individual Assignment #2 - Dynamic Event Registration System ("Coding Conf")

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1305116

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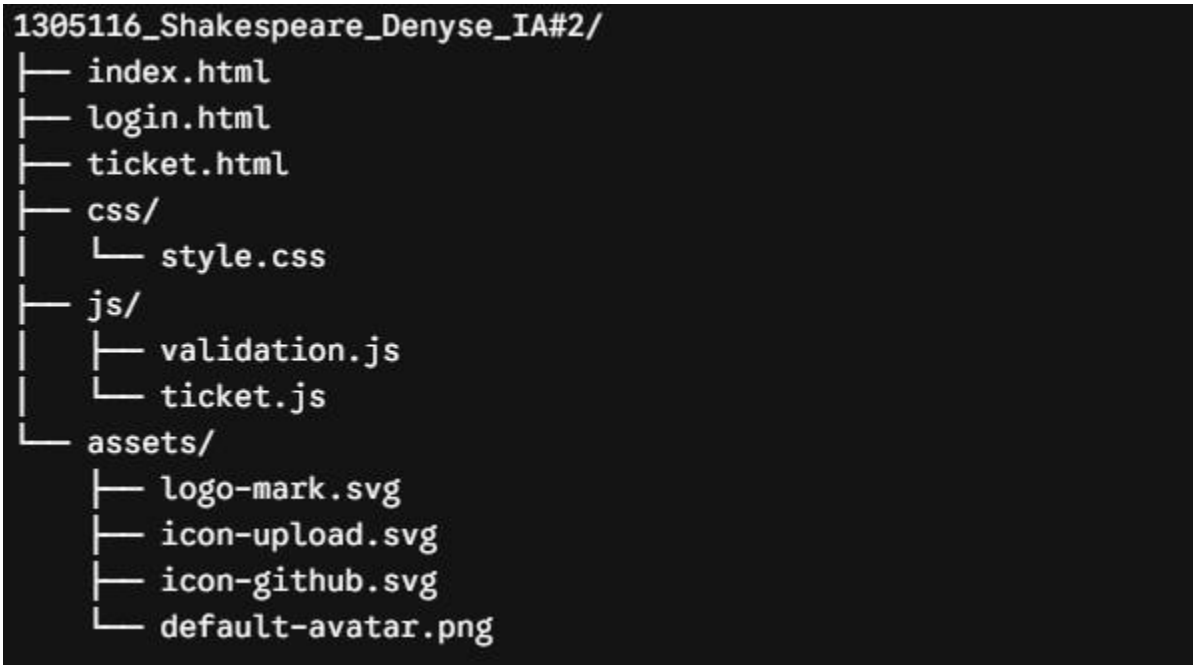
System Overview & Architecture

The Coding Conf 2027 Registration System is a responsive, multi-page client-side web application built using native HTML5, modern CSS3 (Flexbox/Grid, custom properties, fluid typography clamp(), and modern pseudo-classes), and Vanilla JavaScript with localStorage state persistence.

View States Architecture

1. Landing Page (index.html): Displays event promotional headers, schedule metadata (Date: February 5, 2027; Location: Papine, Kgn), structural overview text, and a prominent call-to-action button routing to the registration form.
2. Registration Page (login.html): Captures user inputs (Full Name, Email Address, GitHub Username, and Profile Picture Avatar). Implements client-side regular expression patterns, type/size validation constraints, an interactive drag-and-drop file upload zone (icon-upload.svg) featuring live thumbnail previewing and a dynamic "Remove" option to reset or swap files, alongside localized error messages.
3. Ticket Confirmation Page (ticket.html): Renders a personalized conference pass with a glassmorphism theme, authentic ticket notch cutouts, perforated dashed separator line, serial tracking number, GitHub handle icon (icon-github.svg), and automated route protection.

Workspace Directory Tree & Verified Asset Inventory

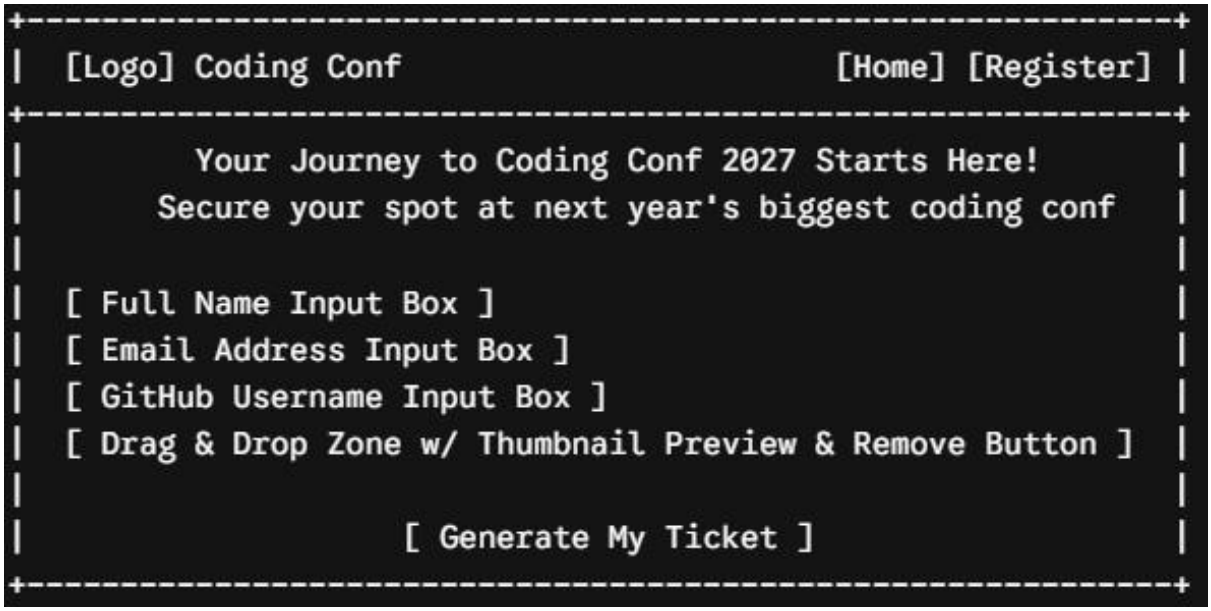


Wireframe & Visual Layout Specifications

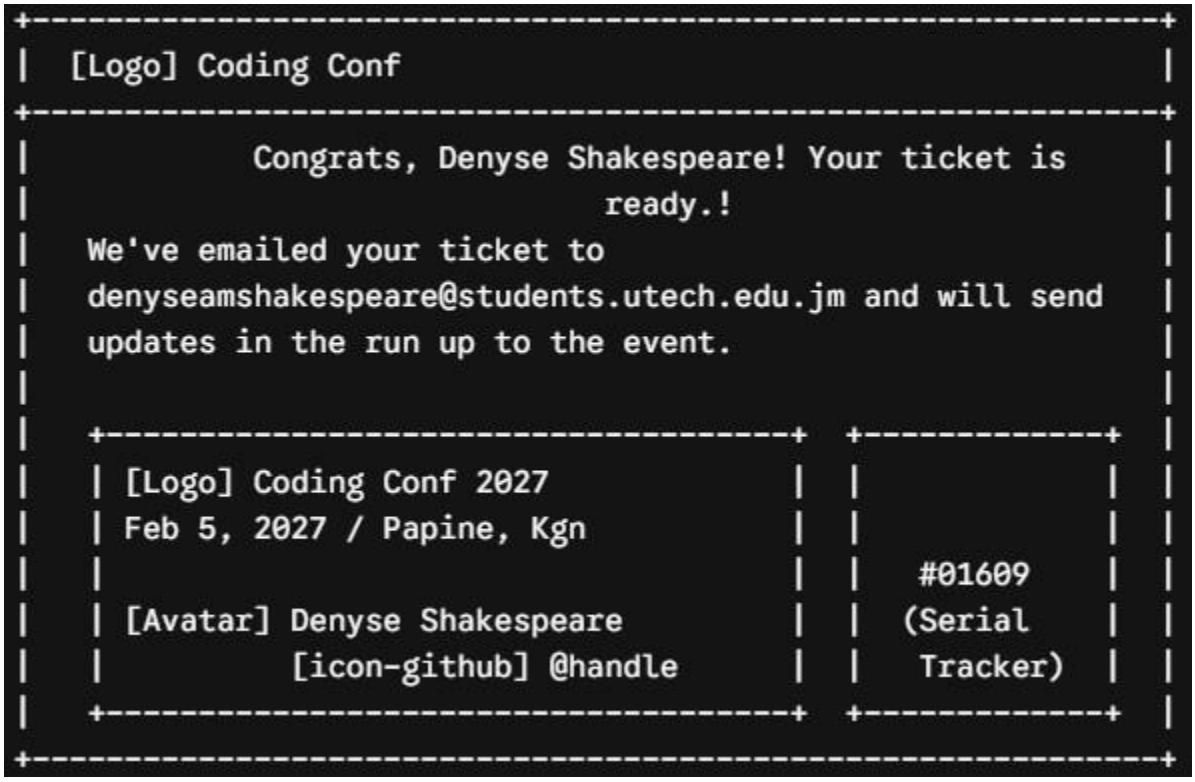
1. Page 1: Landing Page (index.html) Layout Block



2. Page 2: Registration Page (login.html) Layout Block



3. Page 3: Ticket Confirmation Page (ticket.html) Layout Block



Sample Testing Credentials & Validation Rules

To verify form handling and dynamic DOM rendering, the following test case was executed:

Field Name	Test Data Input	Validation Constraint / Rule	Outcome
Full Name	Denyse Shakespeare	Non-empty string check; strips whitespace via .trim()	**Passed**
Email Address	denyseamshakespeare@students.utech.edu.jm	Strict Regex Pattern matching ^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$	**Passed**
GitHub Handle	@Denyse-Shakespeare	Enforces mandatory @ prefix string constraint	**Passed**
Avatar File	profile.png (245 KB)	File type image/jpeg or image/png; size $\le 500\text{ KB}$	**Passed**
Route Security	Direct access to ticket.html	Checks localStorage.getItem('codingConfAttendee'); redirects to index.html if empty	**Passed**

Responsive Design & Accessibility Proof

Responsive Layout Strategy

1. Flexible Units & Fluid Typography: Utilized CSS clamp() functions for heading scaling across screen widths.
2. Layout Containers: Flexbox for headers and metadata lists; CSS Grid for ticket card layouts and media queries at breakpoint 768px and mobile adaptations at 480px.
3. Modern CSS Selectors: Integrated structural functional pseudo-classes :is(), :not(), and :has().

Figure 1: Mobile Viewport Test (~412px width - Pixel 9)



Figure 2: Tablet Viewport Test (~768px width – iPad Mini)

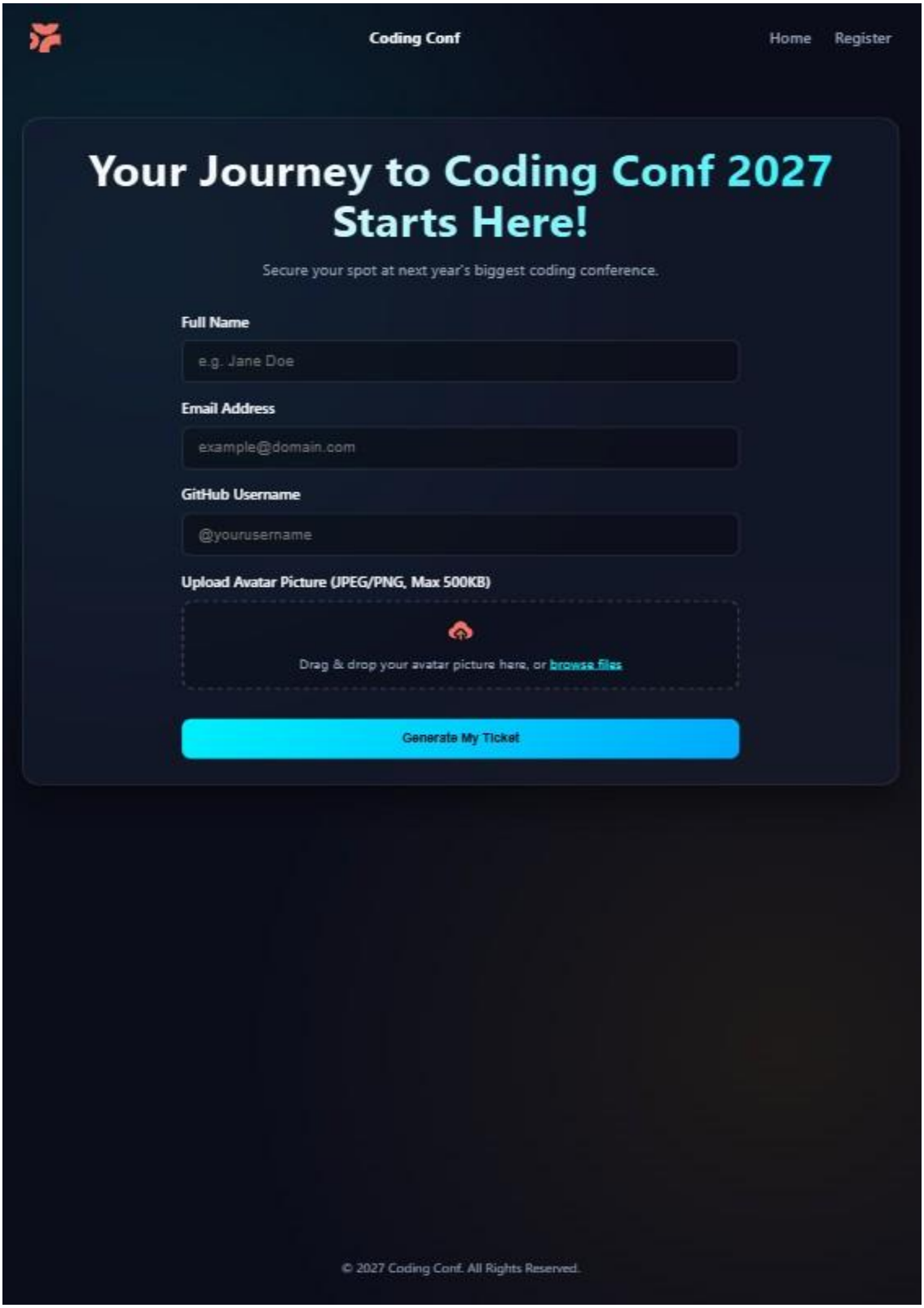
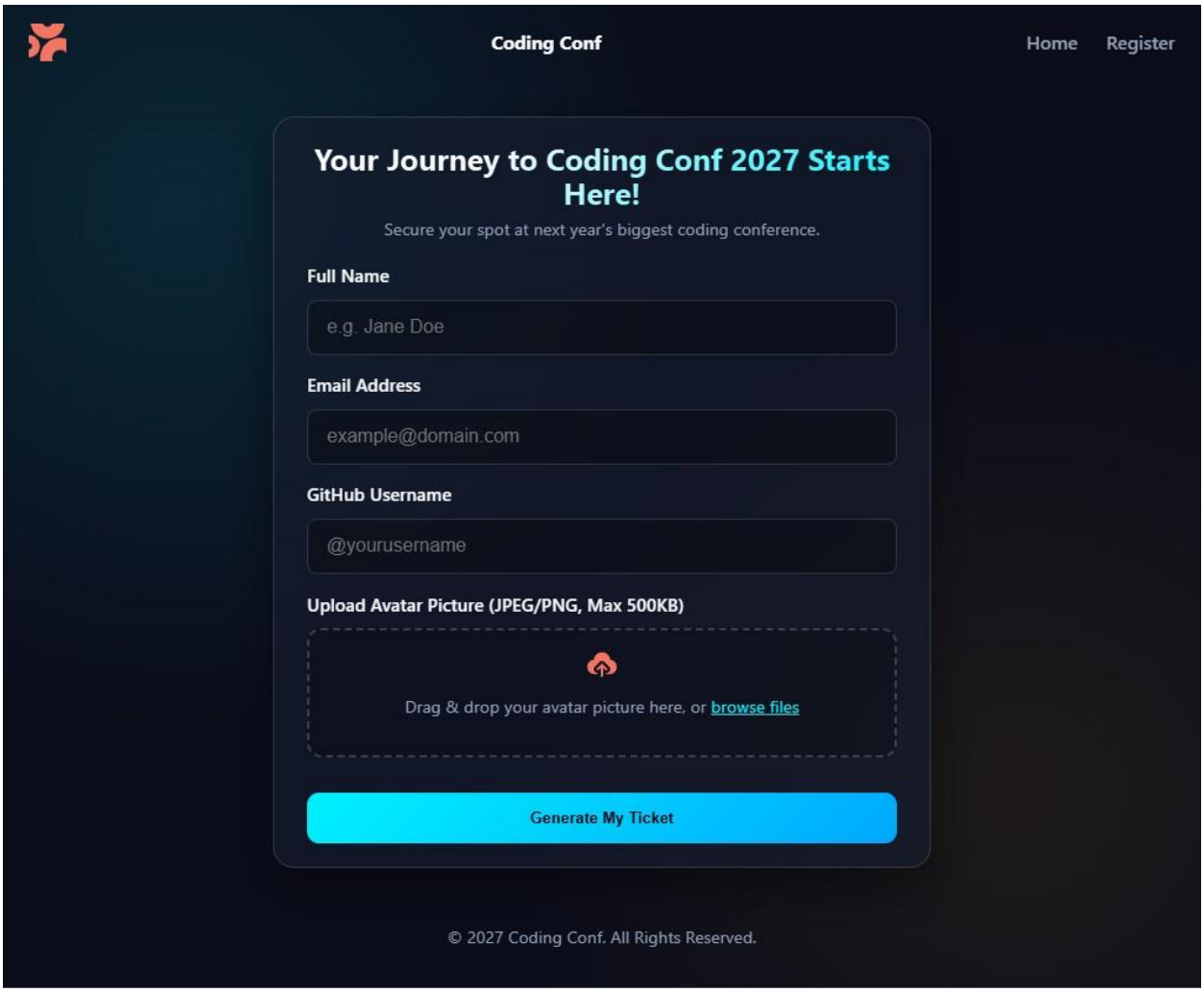


Figure 3: Desktop Viewport Test (1024px+ width)



Deployment Links & Repository Details

GitHub Repository: https://github.com/Denyse-Shakespeare/1305116_Shakespeare_Denyse_IA2/

Live Hosting Link: https://denyse-shakespeare.github.io/1305116_Shakespeare_Denyse_IA2/

AI Disclosure Block

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SCIT Web Programming - AI DISCLOSURE BLOCK

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File Name: index.html / login.html / ticket.html / validation.js

Student Name: Denyse Shakespeare

Student ID: 1305116

Assignment: CIT2011 IA#2: Dynamic Event Registration System

Date: 11/7/2026

GENERIC AI TOOL UTILISED (Check all that apply):

[ ] GitHub Copilot [ ] ChatGPT [ ] Claude [X] Gemini [ ] None

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SCOPE OF WORK & DEBUGGING LOG:

1. Specific Bug/Error Encountered: The native drag-and-drop file upload zone was not cleanly updating the file input object list when dropping images via drag events, causing validation bypasses or incorrect visual feedback.
2. Exact Prompt Sent to AI Tool: "How do I bind a custom drag-and-drop container to a native hidden file input in vanilla JavaScript so that dropping an image updates fileInputElement.files and displays the file name?"
3. Resolution/Fix Applied: Implemented a wrapper event handler in `validation.js` that intercepts 'drop' events, assigns `e.dataTransfer.files` to `fileInputElement.files`, and dynamically updates the drop zone label text while preserving validation rules for JPEG/PNG file types and 500KB size constraints.

⚠ INTEGRITY VERIFICATION DECLARATION:

By submitting this file, I certify that the core algorithmic logic, program architecture, and data structures implemented herein are entirely my own authentic work. AI assistance was restricted strictly to debugging and/or generating explanatory documentation comments in full compliance with the course syllabus.

Student Electronic Signature: D.S.